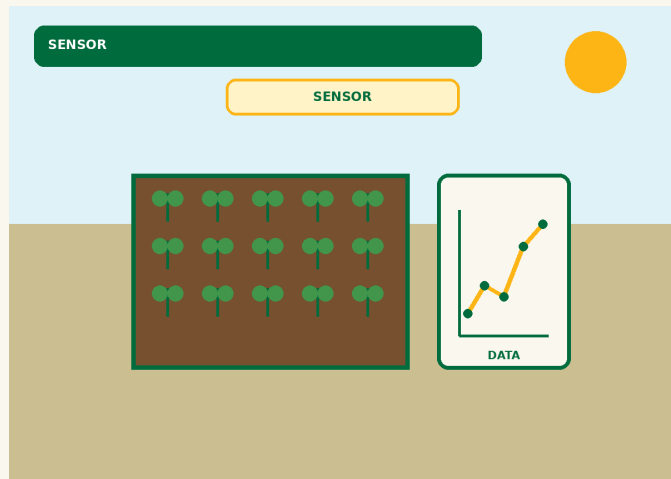




SENSOR

SENSOR

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

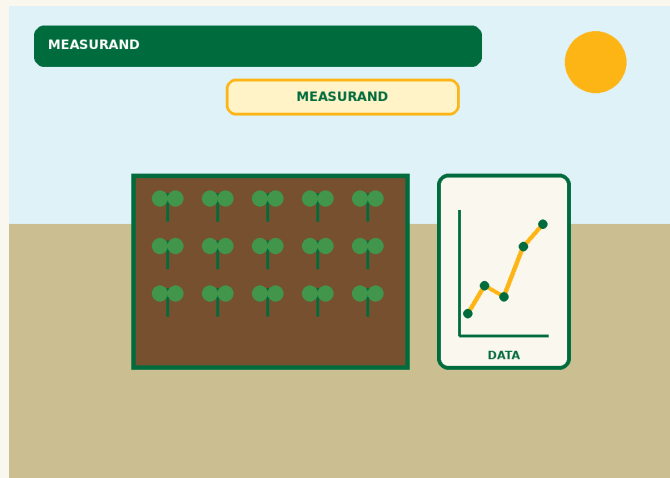
CONCEPT

A sensor converts a physical, chemical, or biological condition into a measurable signal or reading.

INVESTIGATION TASK

Identify the variable measured by each garden sensor.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



MEASURAND

MEASURAND

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 762 • MEASURAND

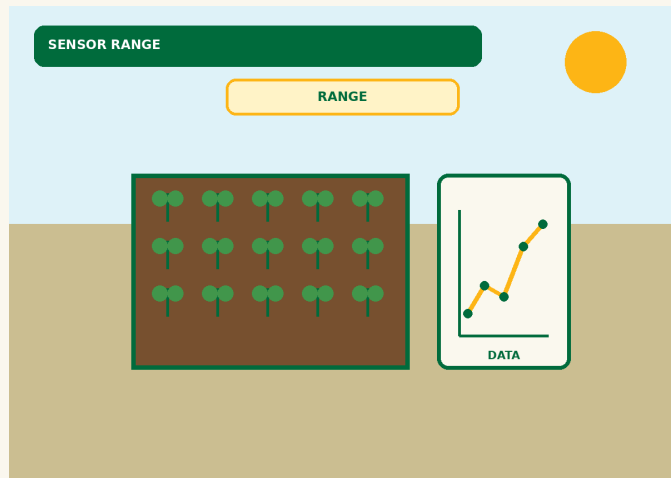
CONCEPT

The measurand is the specific quantity intended to be measured, such as soil moisture, temperature, or water level.

INVESTIGATION TASK

Define the measurand for one sensor.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



RANGE

SENSOR RANGE

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 763 • SENSOR RANGE

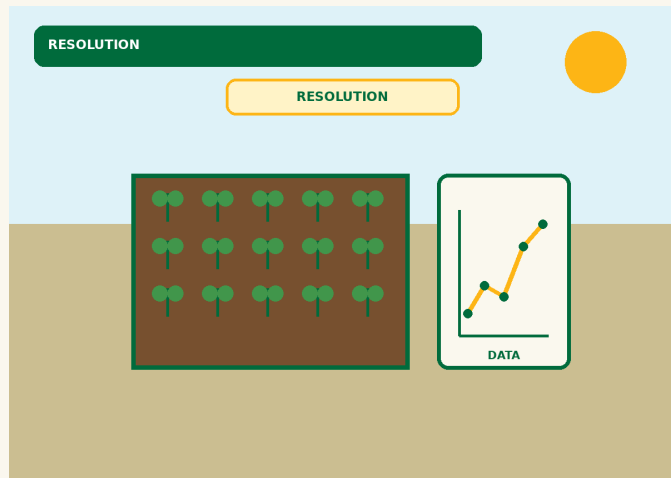
CONCEPT

Sensor range is the interval of values over which an instrument is designed to measure.

INVESTIGATION TASK

Compare expected garden values with sensor range.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



RESOLUTION

RESOLUTION

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 764 • RESOLUTION

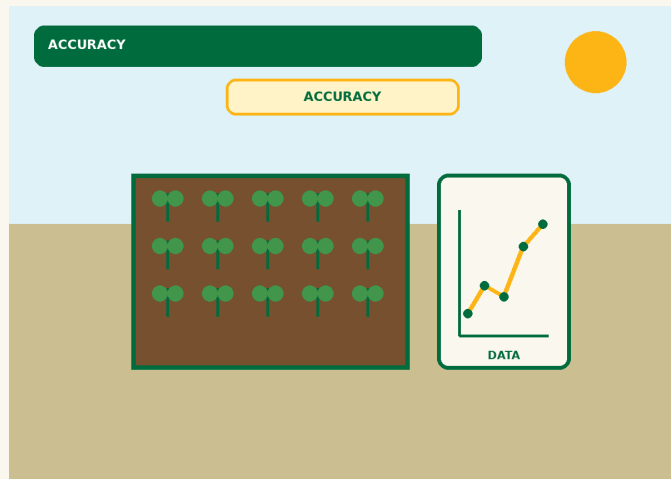
CONCEPT

Resolution is the smallest change in a measured quantity that an instrument can distinguish or report.

INVESTIGATION TASK

Compare the resolution of two instruments.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



ACCURACY

ACCURACY

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 765 • ACCURACY

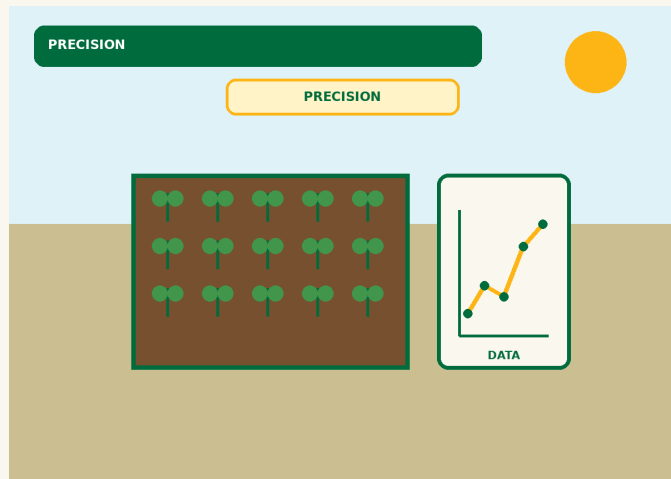
CONCEPT

Accuracy describes closeness of a measurement to an accepted reference value.

INVESTIGATION TASK

Compare a sensor reading with a reference measurement.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



PRECISION

PRECISION

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

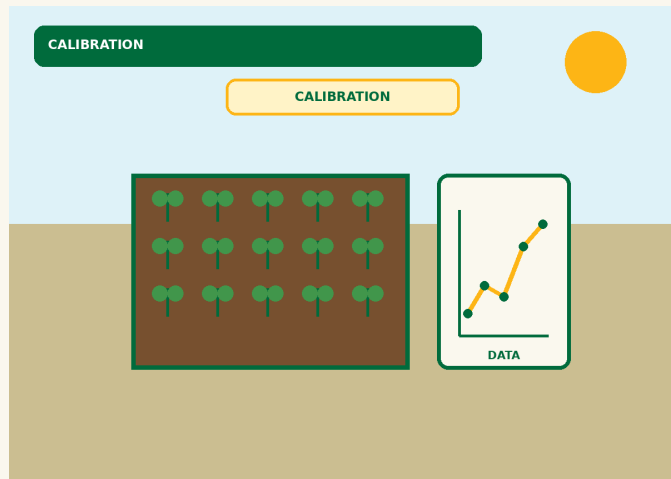
CONCEPT

Precision describes how closely repeated measurements agree with one another.

INVESTIGATION TASK

Evaluate a set of repeated readings.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



CALIBRATION

CALIBRATION

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 767 • CALIBRATION

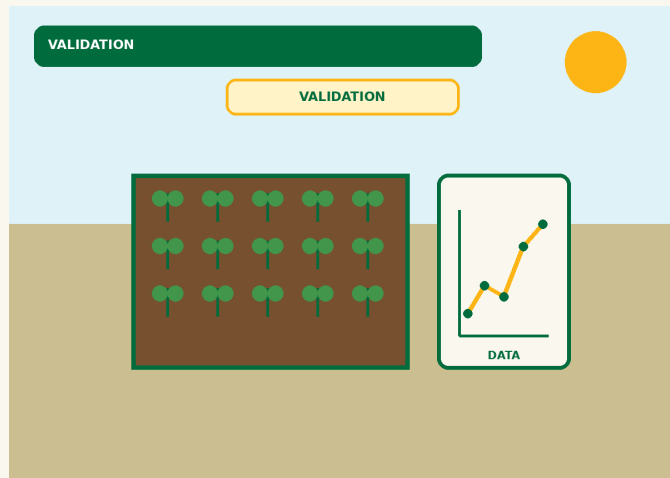
CONCEPT

Calibration establishes the relationship between sensor output and known reference values.

INVESTIGATION TASK

Observe or perform a teacher-guided calibration check.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



VALIDATION

VALIDATION

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 768 • VALIDATION

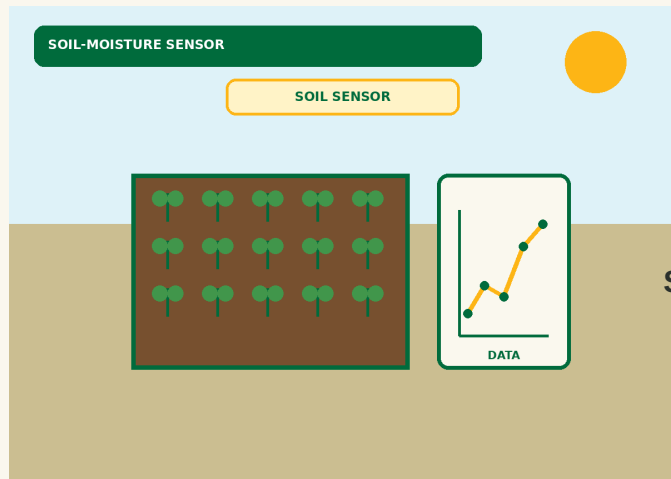
CONCEPT

Validation checks whether sensor measurements remain credible under actual operating conditions.

INVESTIGATION TASK

Compare sensor data with an independent observation.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



SOIL SENSOR

SOIL-MOISTURE SENSOR

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 769 • SOIL-MOISTURE SENSOR

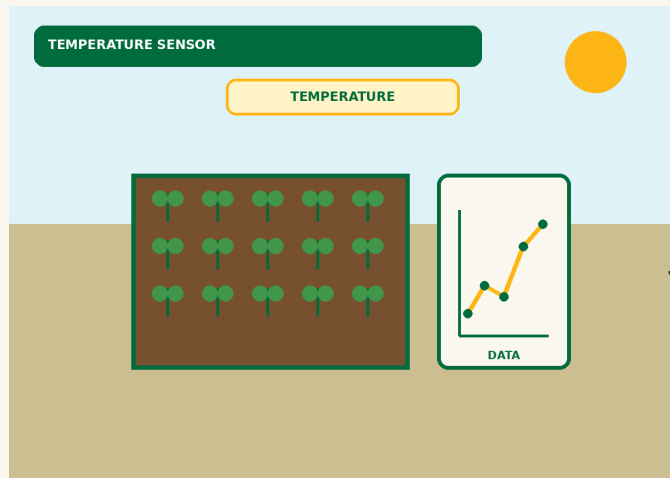
CONCEPT

A soil-moisture sensor estimates a property related to water content or water status in the root zone.

INVESTIGATION TASK

Compare readings at two locations.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



TEMPERATURE

TEMPERATURE SENSOR

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 770 • TEMPERATURE SENSOR

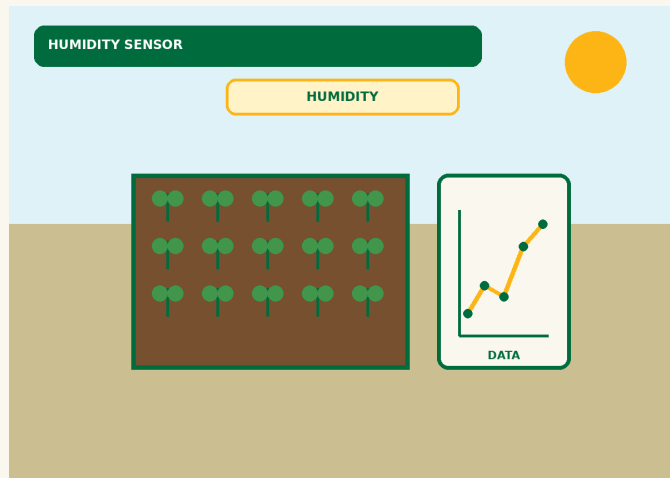
CONCEPT

Temperature sensors measure thermal conditions in air, soil, water, or plant environments.

INVESTIGATION TASK

Record temperatures at defined positions.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



HUMIDITY

HUMIDITY SENSOR

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 771 • HUMIDITY SENSOR

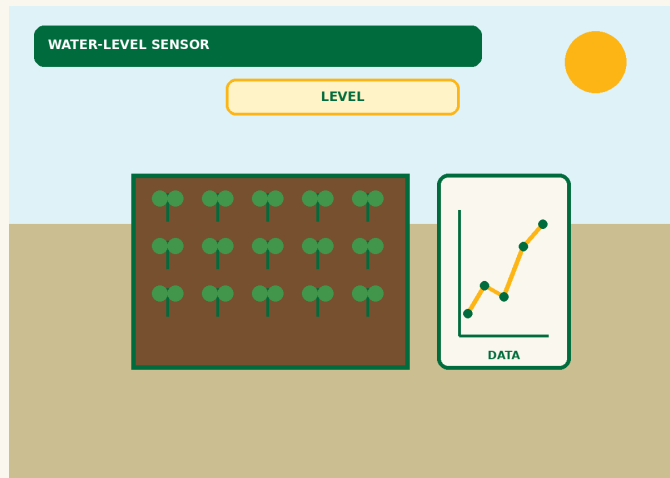
CONCEPT

Relative-humidity sensors measure the amount of water vapor in air relative to saturation at the same temperature.

INVESTIGATION TASK

Compare readings at two times.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



LEVEL

WATER-LEVEL SENSOR

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 772 • WATER-LEVEL SENSOR

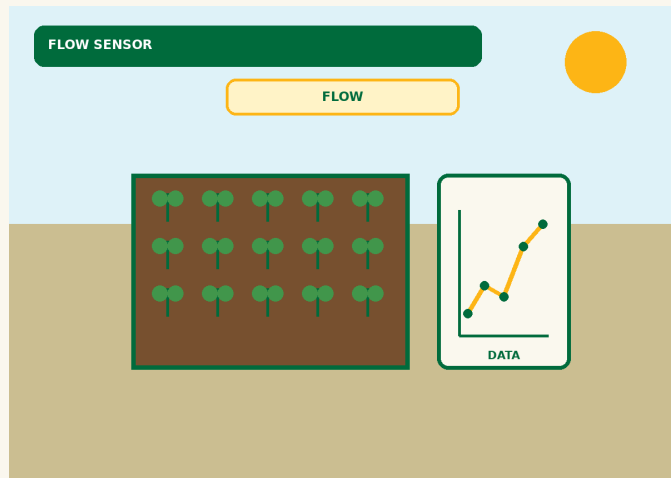
CONCEPT

Water-level sensors estimate the height or quantity of water in a tank or reservoir.

INVESTIGATION TASK

Compare the sensor reading with a visual scale.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



FLOW

FLOW SENSOR

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 773 • FLOW SENSOR

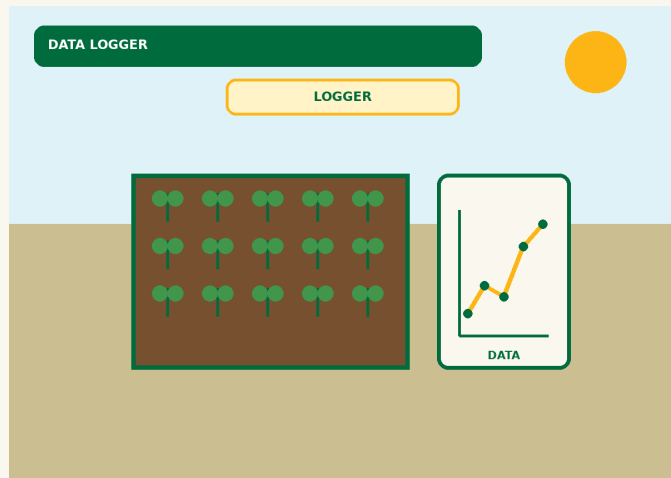
CONCEPT

Flow sensors measure the rate or accumulated volume of water moving through a pipe or channel.

INVESTIGATION TASK

Compare meter readings during irrigation.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



LOGGER

DATA LOGGER

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 774 • DATA LOGGER

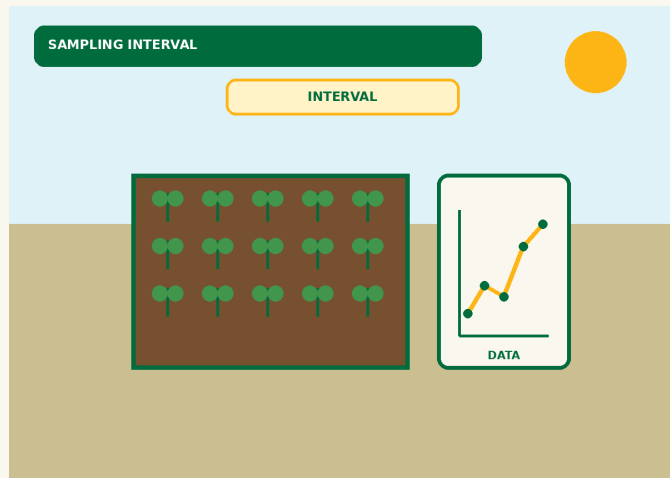
CONCEPT

A data logger stores time-stamped sensor readings for later analysis.

INVESTIGATION TASK

Inspect a time series from one sensor.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



INTERVAL

SAMPLING INTERVAL

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 775 • SAMPLING INTERVAL

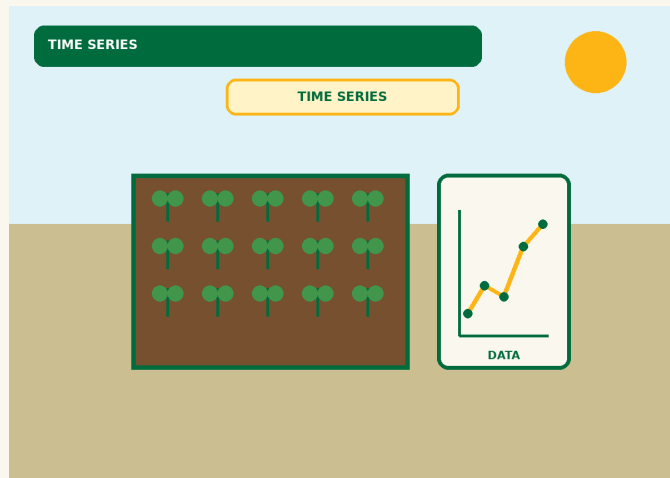
CONCEPT

Sampling interval is the time between consecutive measurements recorded by a monitoring system.

INVESTIGATION TASK

Compare 1-minute and 30-minute datasets.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



TIME SERIES

TIME SERIES

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

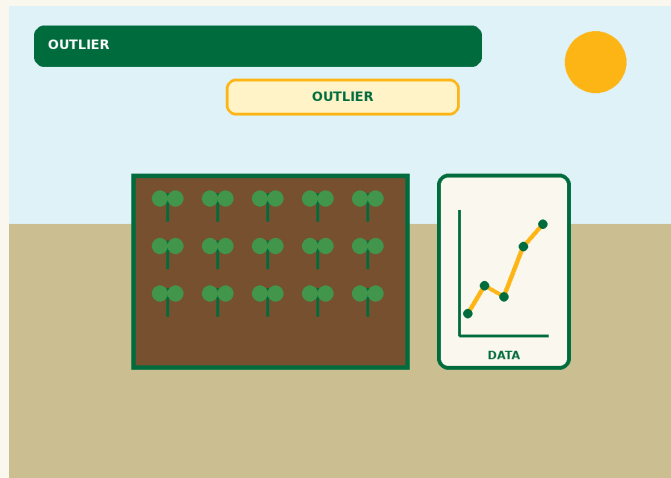
CONCEPT

A time series is a sequence of measurements ordered by time.

INVESTIGATION TASK

Identify a trend and an abrupt change.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



OUTLIER

OUTLIER

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 777 • OUTLIER

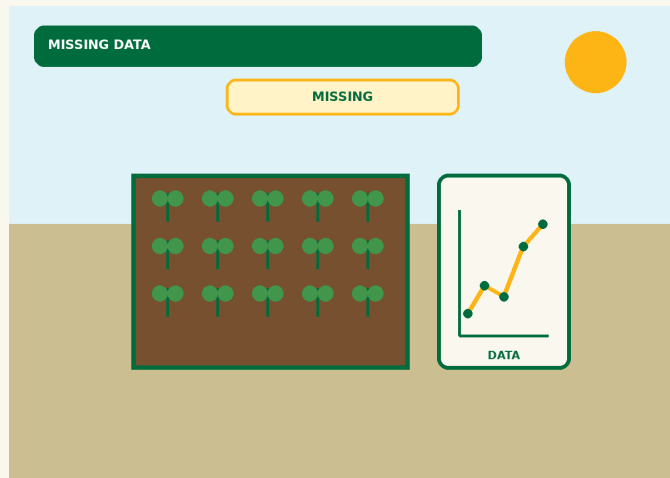
CONCEPT

An outlier is an observation that differs strongly from surrounding or expected values and requires investigation rather than automatic deletion.

INVESTIGATION TASK

Flag and investigate one suspicious value.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



MISSING

MISSING DATA

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 778 • MISSING DATA

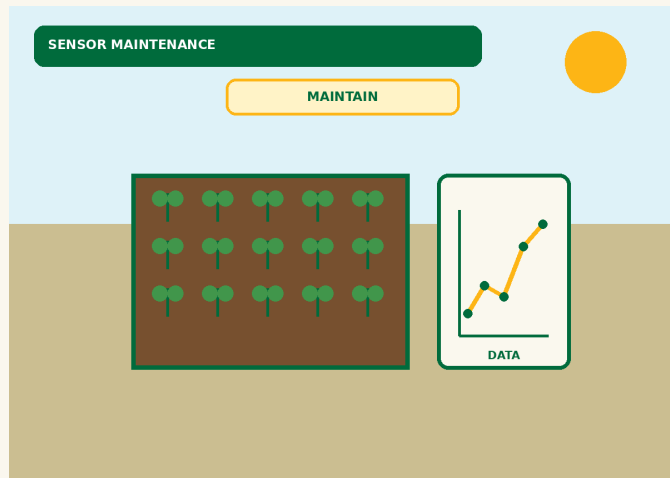
CONCEPT

Missing data are expected observations that were not recorded because of communication, power, sensor, or logging problems.

INVESTIGATION TASK

Identify gaps in a time series.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



MAINTAIN

SENSOR MAINTENANCE

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 779 • SENSOR MAINTENANCE

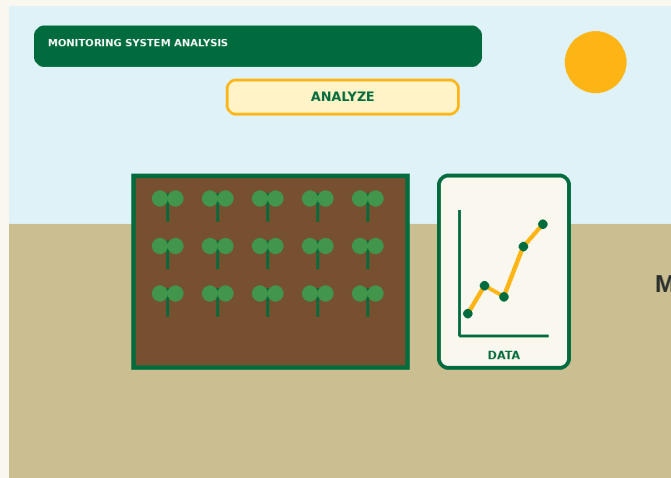
CONCEPT

Reliable monitoring requires inspection, cleaning, calibration checks, secure connections, and documented maintenance.

INVESTIGATION TASK

Create a maintenance schedule.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



ANALYZE

MONITORING SYSTEM ANALYSIS

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 780 • MONITORING SYSTEM ANALYSIS

CONCEPT

Monitoring-system analysis considers sensor suitability, data quality, placement, calibration, timing, and decision usefulness.

INVESTIGATION TASK

Evaluate one sensor installation and recommend an improvement.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE